

$\mathcal{H}_1$ AND $\mathcal{H}_2$ DO NOT CHARACTERIZE MÖBIUS UNCERTAINTY

ABSTRACT. Sahay left open, while writing that he considered it “quite unlikely”, whether the Möbius uncertainty property $\mathcal{M}$ of a locally finite poset with a minimum is characterized by the two necessary conditions $\mathcal{H}_1$ and $\mathcal{H}_2$. We confirm his expectation: the characterization fails. For every $r \geq 1$, we construct a countable locally finite poset P_r with a minimum such that P_r satisfies $\mathcal{H}_k$ exactly for $1 \leq k \leq r$, but fails $\mathcal{M}$. Thus no finite family of the conditions $\mathcal{H}_k$ implies Möbius uncertainty.

Let P be a locally finite poset with minimum $\hat{0}$. For $f: P \rightarrow \mathbb{C}$, define

$$(Zf)(x) = \sum_{y \leq x} f(y), \quad \text{supp}(f) = \{x \in P : f(x) \neq 0\}.$$

The poset P has the *Möbius uncertainty property* $\mathcal{M}$ if, for every nonzero f , at least one of $\text{supp}(f)$ and $\text{supp}(Zf)$ is infinite. For $k \geq 1$, it has property $\mathcal{H}_k$ if every k -element set $S \subseteq P$ contains some $z \in S$ for which there are infinitely many $x \in P$ such that

$$x \geq z \quad \text{and} \quad x \not\geq y \quad (y \in S \setminus \{z\}).$$

Pollack proved the arithmetic uncertainty principle [2], and Goh formulated and studied its poset analogue [1]. Sahay introduced the properties $\mathcal{H}_k$ and proved

$$\mathcal{H}_{\mathbb{N}} \implies \mathcal{M} \implies \mathcal{H}_1 \text{ and } \mathcal{H}_2,$$

where $\mathcal{H}_{\mathbb{N}}$ means that every $\mathcal{H}_k$ holds [3, Theorems 1.2 and 1.5]. After showing that $\mathcal{M} \not\Rightarrow \mathcal{H}_3$ [3, Theorem 1.11], he wrote, immediately following that theorem [3, p. 5]:

“This leaves open the possibility that

$$P \text{ has } \mathcal{M} \iff P \text{ has } \mathcal{H}_1 \text{ and } \mathcal{H}_2,$$

for general posets, but I consider this quite unlikely.”

Theorem 1. *For every integer $r \geq 1$, there is a countable locally finite poset P_r with a minimum such that*

$$P_r \text{ has } \mathcal{H}_k \iff 1 \leq k \leq r, \quad P_r \text{ does not have } \mathcal{M}.$$

Proof. Let $\mathcal{F}(\mathbb{N})$ be the poset of finite subsets of $\mathbb{N} = \{1, 2, \dots\}$ ordered by inclusion. For $1 \leq i \leq r$, let

$$Q_i = \{A_i : A \in \mathcal{F}(\mathbb{N})\}$$

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be a copy of this poset. Define

$$P_r = \{\widehat{0}\} \sqcup \bigsqcup_{i=1}^r Q_i$$

by placing $\widehat{0}$ below every element, retaining inclusion within each Q_i , and imposing no comparabilities between distinct copies. The poset is countable and locally finite, since

$$[A_i, B_i] = \{C_i : A \subseteq C \subseteq B\}, \quad [\widehat{0}, B_i] = \{\widehat{0}\} \cup \{C_i : C \subseteq B\}.$$

Fix $1 \leq k \leq r$ and let $S \subseteq P_r$ have size k . If $\widehat{0} \in S$, then the other $k-1$ elements occupy at most $r-1$ copies. Choose a copy Q_i disjoint from $S \setminus \{\widehat{0}\}$. Taking $z = \widehat{0}$, every element of Q_i is an exclusive upper witness for z .

Suppose $\widehat{0} \notin S$. Choose an occupied copy Q_i , put $T = S \cap Q_i$, and choose $A_i \in T$ minimal under inclusion. For every $B_i \in T \setminus \{A_i\}$, choose $b_B \in B \setminus A$, and let

$$E = \{b_B : B_i \in T \setminus \{A_i\}\}.$$

For each $c \in \mathbb{N} \setminus (A \cup E)$, set $X_c = (A \cup \{c\})_i$. Then $X_c \geq A_i$, while $B_i \not\leq X_c$ for every $B_i \in T \setminus \{A_i\}$; elements of $S \setminus T$ lie in other copies and are incomparable with X_c . Since there are infinitely many choices of c , this proves $\mathcal{H}_k$.

Now let $k \geq r+1$ and extend

$$S_0 = \{\widehat{0}, \emptyset_1, \dots, \emptyset_r\}$$

to a k -element set $S \subseteq P_r$. For $z = \widehat{0}$, the only element above z that is not above any other element of S is $\widehat{0}$ itself, since every nonminimum element lies above some $\emptyset_i$. For $z \neq \widehat{0}$, every element above z is also above $\widehat{0}$. Thus S has no element with infinitely many exclusive upper witnesses, and P_r fails $\mathcal{H}_k$.

Finally, define $f: P_r \rightarrow \mathbb{C}$ by

$$f(\widehat{0}) = 1, \quad f(\emptyset_i) = -1 \quad (1 \leq i \leq r), \quad f(x) = 0 \quad \text{otherwise.}$$

For every $A_i \in Q_i$,

$$(Zf)(A_i) = f(\widehat{0}) + f(\emptyset_i) = 0, \quad (Zf)(\widehat{0}) = 1.$$

Hence both f and Zf are nonzero and finitely supported, so P_r does not have $\mathcal{M}$. $\square$

Corollary 2. *The possibility left open in [3, p. 5] is false. More generally, no finite collection of the conditions $\mathcal{H}_k$ implies the Möbius uncertainty property for locally finite posets with a minimum.*

Proof. The first assertion is Theorem 1 with $r = 2$. Given a finite set $K \subseteq \mathbb{N}$, take $r \geq \max(K \cup \{1\})$; then P_r has every $\mathcal{H}_k$ with $k \in K$ but does not have $\mathcal{M}$. $\square$

Remark 3. The displayed witness is support-minimal. Indeed, if $0 \neq h: P_r \rightarrow \mathbb{C}$ has $s = |\text{supp}(h)| \leq r$, then $\mathcal{H}_s$ supplies infinitely many x for which $(Zh)(x) = h(z) \neq 0$ for some $z \in \text{supp}(h)$. Thus every finitely supported witness to failure of $\mathcal{M}$ has support at least $r + 1$, and the function f above attains this bound.

In P_r the copies $Q_1, \dots, Q_r$ are the up-sets of $\emptyset_1, \dots, \emptyset_r$ and partition $P_r \setminus \{\widehat{0}\}$. What makes $\mathcal{H}_1, \dots, \mathcal{H}_r$ survive is the number of copies rather than the partition alone: for $\widehat{0} \in S$ it is the pigeonhole over exactly r copies that leaves a copy free, while for $\widehat{0} \notin S$ the witnesses are produced inside a single copy, for a set of any size. Among the sets of size $r + 1$, the only one with no element having infinitely many exclusive upper witnesses is $S_0 = \{\widehat{0}, \emptyset_1, \dots, \emptyset_r\}$, and the failure of $\mathcal{H}_k$ for $k \geq r + 1$ comes from the extensions of S_0 . For $r \geq 2$, the poset P_r is not a join-semilattice, since elements in distinct copies have no common upper bound; hence the construction lies outside Sahay's join-semilattice theorem [3, Theorem 1.10 and Section 4].

REFERENCES

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